



The Power of Words: Gender Priming and Financial Choices

Project for Irrationality, Economics and Finance: A behavioural and experimental approach

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Motivation and Research Questions:

In recent years, families and small investors have been facing new and more complex financial instruments. In this context, small, non fundamental variations in the beliefs can change significantly the demand for risky assets, especially for first time investors. The goal of this project is to understand whether a brief text that underlines behavioural differences between genders can change propensity to risk.

One of the most widely documented behavioural regularities relates to the difference between gender in risk taking and investment behaviour; a large empirical literature shows that women tend to invest less in financial products, and on average, choose less risky assets. We wonder whether these differences in risk taking preferences is also perceived by individuals, in this case CBS students, and whether making gender difference salient can shift their financial choices. In other words, we investigate whether a short priming statement about gender differences can alter beliefs and individual behaviour.

Our contribution to the literature is the following: we want to check if this patterns, documented in previous papers, is consistent and significant within a university environment, focusing on students of economics and finance. We aim to accomplish this by constructing two types of indexes; the first one assesses perceptions about gender differences (belief index) and the second one measures risk aversion (risky share). These indexes are derived from a dataset constructed through Qualtrics that contains multiple questions about demographics (eg. age and sex), beliefs and investment experience; students were asked to assign weights to a hypothetical portfolio and then, if they invest, to describe their portfolio composition (more details in data & survey section). The results of this paper could have relevant policy implications:

Rather than reflecting these differences, governments could use public policies, such as incentives or subsidies for the design of the financial products and financial communication, to address this asymmetry. Our research questions are the following:

Does gender have a relevant effect on risk aversion for educated students?

Do people believe that there is actually a difference in risk aversion based on gender?

Literature review:

A large empirical literature documents gender differences in financial risk taking and portfolio choice. What do we know so far? Across surveys, experimental designs, administrative records, women consistently reported a higher risk aversion compared to men (less likely to invest in risky assets). These findings are consistent not only in different sub groups but also in different countries and cultures, suggesting that gender has an important relevance in terms of investing behaviour.

Lusardi & Mitchell¹ show that financial literacy has an important impact in investment decisions. Highly educated people are more likely to invest in financial markets, to plan for their retirement and tolerate higher levels of risk in their portfolios. They analyze several demographic characteristics and conclude that there is a positive correlation between age and stock holdings (the older you are the more stocks you are likely to own); similarly, they find that financially educated people are more likely to own stocks. Income and marriage status are also positively related to stock ownership. Similar conclusions are reached

by Hilgert, Hogarth and Beverly, who show that there is a correlation between financial knowledge and investment behaviour. Van Rooij, Kool and Prast further document that risk return preferences are associated with financial literacy and that financially educated individuals are better able to make informed portfolio decisions. These findings imply that this difference between genders in financial investments may be due to difference in education.

Another important contribution comes from research on survey risk preferences. Barsky, Juster, Kimball and Shapiro show that hypothetical questions on risk tolerance possess strong behavioral validity: individuals' stated preferences are highly correlated with their actual portfolio choices.

Hibbert et al. show that financial knowledge and financial literacy appear to mediate the gender gap in risk attitudes. Using U.S. survey based data, they find that higher financial knowledge is associated with greater risk tolerance, reducing the gender differential; in their regressions, the coefficient on female is negative, indicating that women are, on average, more risk averse.

Wu documents that gender differences in preferences map directly into differences in actual portfolio allocations: women often choose less risky portfolios compared to men. Similarly, van Rooij, Lusardi and Alessie find that individuals with higher financial literacy are more likely to participate in the stock market. This effect is also driven and influenced by gender.

All these studies are conducted in the USA, Netherlands, Finland and more generally, Northern Europe. In summary, there are many other papers with similar findings, so we can conclude that across several countries women exhibit more risk averse behavior than men, not only in surveys but also in administrative data. This motivates our work: we test whether the same pattern holds among women with high financial literacy in an economics university environment.

Data & Survey:

Data construction was the most challenging part of the project. We first clarified the research question and the indices required to test our hypotheses, and then designed a survey (see next page) to collect the necessary variables such as sex, hypothetical portfolio composition and actual portfolio composition, investing experience, overconfidence and education. We use this data to construct two indexes that will have an important role in defining risk aversion and beliefs (see theoretical framework section).

The survey consists of six parts:

Section A: We ask questions about perceptions of risk taking. Responses are on a 1 to 7 scale, where 1 = “strongly disagree” and 7 = “strongly agree”.

Section B: We ask whether you currently invest and how many years of investing experience you have. We then ask to build a hypothetical portfolio by allocating weights across four asset classes where some assets are riskier than others. The total must sum to 100%.

Section C: We ask economics/finance questions to measure financial literacy, each with five answer options.

Section D: We ask how many financial literacy questions you believe you answered correctly, and what your expected return on your investments would be.

Section E: If you indicated that you currently invest, we ask for the composition of your actual portfolio.

Section F: We ask about your education and your gender.

Here you can overview the survey questions and potential answers:

Behavioral Finance — Survey							
A. Perceptions about risk-taking (1=Strongly disagree, 7=Strongly agree)							
Men and women tend to differ in their willingness to take financial risks.	1	2	3	4	5	6	7
Younger individuals are generally more willing to take financial risks than older individuals.							
I believe investment behavior is similar across genders.							
People with higher education are generally more willing to take financial risks.							
Differences in risk-taking between genders are mostly due to personality rather than financial knowledge.							
C. Financial knowledge							
Rates & duration: If market yields rise by 1%, which 5-year bond loses more value?							
a.) 5-year zero-coupon b.) 5-year 5% coupon c.) Same d.) Cannot determine e.) I'm uncertain about this							
If the interest rate falls, what should happen to bond prices?							
a.) Rise b.) Fall c.) Stay the same d.) None of the above e.) I'm uncertain about this							
The term premium on long-term bonds compensates investors for							
a.) Short-term rate volatility & inflation uncertainty b.) Credit risk c.) Default risk d.) Tax differences between bonds e.) I'm uncertain about this							
Options: If underlying volatility rises, the value of a call option will							
a.) Increase b.) Decrease c.) Stays the same d.) Only change at maturity e.) I'm uncertain about this							
When the central bank conducts quantitative easing (QE), which is correct?							
a.) Raises short-term interest rates b.) Sells government bonds to reduce liquidity c.) Tightens reserve requirements d.) Purchases long-term securities to lower yields and stimulate spending e.) I'm uncertain about this							
B. Investment background							
Do you engage in investments (bonds, mutual funds, ETFs, stocks, cryptocurrencies)?							
a.) Yes b.) No							
Years of investing experience (if applicable): _____ years							
Hypothetical allocation (sum=100%):							
Savings/fixed deposits _____% Bonds _____% Mutual funds/ETFs _____%							
Stocks/cryptocurrencies _____% Total _____%							
D. Confidence & expectations							
How many knowledge questions did you answer correctly above?							
a.) 0 b.) 1 c.) 2 d.) 3 e.) 4 f.) 5							
Your expected annual return next year: _____%							
E. Actual portfolio (sum=100%)							
Savings/fixed deposits _____% Bonds _____% Mutual funds/ETFs _____%							
Stocks/cryptocurrencies _____% Others (e.g., housing) _____% Total _____%							
F. Demographics							
Gender: a.) Male b.) Female c.) Others							
Highest education:							
a.) Bachelor—First yr b.) Bachelor—Second yr c.) Bachelor—Third yr d.) Bachelor—Fourth yr e.) Bachelor—Graduated f.) Masters—First yr g.) Masters—Second yr h.) Master—Graduated i.) Doctoral degree (PhD or eq.)							

Identification strategy:

Before the survey, we showed a short sentence: the first part of this sentence (marked with 1) will be the same in both the control and the treatment group, while the second one (marked with 2) is the prime and it appears randomly (people who will see the message are the treatment group, while the others are the control group):

- 1) People approach investment decisions in different ways depending on a variety of factors, including how they view risk, responsibility, and long term planning. These factors are sometimes associated with social roles and identity, such as gender.
Low financial literacy is strongly associated with non-participation in the stock market.
The risky share conditional on participating is fairly flat early on and declines at older ages.
- 2) Survey of lab/field experiments identifies robust gender differences in risk preferences, Women are, on average, more risk-averse.

Our prime makes gender identity salient right before the decision. Noting that men and women often differ in risk taking brings gender differences to mind. With this information, we expect to shift choices toward the risk level people think fits their gender. The survey asks the respondents the same identical questions, however, when we retrieved the data, we were able to distinguish between participants that are part of the treatment group and participants who are part of the control group. This is essential to check whether there is a difference in the behaviour of the two groups.

Theoretical framework and hypothesis:

We built two indexes: the Risk Aversion index and the Beliefs index; we built them as follows:

$$\begin{aligned}
 s(x) &= \frac{x-1}{6}, \quad x \in \{1, \dots, 7\}, \\
 \text{Core: BeliefIndex}_{\text{core}} &= \frac{1}{2}[s(A_1) + 1 - s(A_3)], \\
 \text{Extended: BeliefIndex}_{\text{ext}} &= \frac{1}{3}[s(A_1) + 1 - s(A_3) + s(A_5)].
 \end{aligned}
 \quad
 \begin{aligned}
 \text{Raw (1-7): BeliefIndex}^{(1-7)} &= \frac{A_1 + (8 - A_3) + \lambda A_5}{2 + \lambda}, \\
 \lambda &\in \{0, 1\}, \\
 \text{Map to [0,1]: BeliefIndex}_{[0,1]} &= \frac{\text{BeliefIndex}^{(1-7)} - 1}{6}.
 \end{aligned}$$

$$\begin{aligned}
 r_{\text{Sav}} &= 0, \quad r_{\text{Bonds}} = 0.5, \quad r_{\text{MF/ETF}} = 0.75, \quad r_{\text{Stocks}} = 1. \\
 \text{RiskyShare}^{\text{hyp}} &= \frac{r_{\text{MF/ETF}} w_{\text{MF/ETF}} + r_{\text{Stocks}} w_{\text{Stocks}} + r_{\text{Bonds}} w_{\text{Bonds}}}{100}, \\
 \text{RiskyShare}^{\text{act}} &= \frac{r_{\text{MF/ETF}} w_{\text{MF/ETF}} + r_{\text{Stocks}} w_{\text{Stocks}} + r_{\text{Bonds}} w_{\text{Bonds}} + r_{\text{Others}} w_{\text{Others}}}{100}.
 \end{aligned}$$

Lamda is a dummy set to 0 when we exclude A5 question and 1 when we include it. We constructed two versions of the Belief Index, the core and the extended one. The difference lies in the fact that question s(A5) is included or not. The Risky share indexes are constructed in such a way that riskier assets have a higher weight. This measures are usefull to check whether there is actually a difference in responses of participants who answered the survery as a treatment group or as a control group and they are our dependent variables in the main regressions.

The regressions that we run have the following explicit form:

$$\begin{aligned}
 \text{Belief Index}_i &= \alpha + \beta_1 \text{Female}_i + \beta_2 \text{Gender_Prime}_i + \beta_3 \text{Female}_i \times \text{Gender_Prime}_i \\
 &\quad + \beta_4 \text{Education}_i + \beta_5 \text{Finance Literacy}_i + \beta_6 \text{OC} + \varepsilon_i \\
 \text{RiskLevel}_i &= \alpha + \beta_1 \text{Female}_i + \beta_2 \text{Gender_Prime}_i + \beta_3 \text{Female}_i \times \text{Gender_Prime}_i \\
 &\quad + \beta_4 \text{Education}_i + \beta_5 \text{Finance Literacy}_i + \beta_6 \text{InvestmentHistory}_i \\
 &\quad + \beta_7 \text{OC} + \varepsilon_i
 \end{aligned}$$

Both regressions use the same variables as controls. We are mostly interested in interpreting the coefficients of the following variables: Female, Gender_Prime, Female x Gender_Prime.

Our hypotheses:

RiskLevel: Male participants hold more risky assets $\Rightarrow$ women have lower risky share $\Rightarrow \beta_1 < 0$.

The prime is designed to reduce the gap between the genders.

BeliefIndex: We are not sure about the direction of the Belief Index, since it could be the case that male participants perceive more gender based differences than female participants do, or the opposite. However, we expect convergence of beliefs after the prime.

Data Cleaning and Variable Construction:

We started by exporting the data from Qualtrics (platform used to create the survey); in total the survey was open by 192 students, however we obtained and used only 55 observations to run the regression and for the analysis, since most of these observations in the survey were actually empty or incomplete. After exporting the data we constructed manually the index in excel. In order to create the final dataset for the regression we did the following:

- We collected A1, A3 and A5 and we transformed the string answers into numbers (1 to 7).
- Using excel we computed s(A1), s(A3), s(A5)
- We built the 4 indexes as described above (using excel)
- We transformed Female dummy from string to number (1 for female, 0 for male).
- We transformed education variables into years of education (this way we will use less degrees of freedom to run the regression afterwards)
- We computed the overconfidence with the following formula: we asked participants the expected return for next year: we subtracted the percentage that they chose by 6,75% (close to the average of the market). Note: we also have another measure of overconfidence (in financial literacy) but we will use it later for comparison (we didn't use it in the regression)
- Financial knowledge: we assigned 1 to correct answer and 0 to incorrect. We summed all the values for these answers. We also asked students who participated in the survey how many they think they got correctly (second measure of overconfidence)

In the end, our final dataset consisted of the following variables:

Belief index (core)	Belief index (ext)	Risky share Act	Risky share hyp	Financial lit	Expected correct answers
Educ	Overconfidence	Experience	Prime	Female	

Results:

After running the regression we obtained the following results:

Variable	Coef	StdErr	t	P> t	[95%_low]	[95%_high]
female	-0,1194	0,176162	-0,68	0,503	-0,4802491	0,2414552
prime	0,102566	0,145393	0,71	0,486	-0,1952585	0,4003897
female#prime	-0,13842	0,211358	-0,65	0,518	-0,5713651	0,2945307
educ	-0,02731	0,042102	-0,65	0,522	-0,1135573	0,058928
overconf	-0,00309	0,013297	-0,23	0,818	-0,0303314	0,0241418
exper	-0,00245	0,024687	-0,1	0,922	-0,0530197	0,0481166
finlit	-0,06716	0,056895	-1,18	0,248	-0,1837038	0,0493854
_cons	1,332213	0,709302	1,88	0,071	-0,1207267	2,785153
Statistic	Value					
Dependent var	risky_share_act					
Number of obs	36					
F-statistic	0,69					
Prob > F	0,6816					
R-squared	0,1467					
Adj R-squared	-0,0667					

Variable	Coef	StdErr	t	P> t	[95%_low]	[95%_high]
female	-0,17422	0,088707	-1,96	0,057	-0,3535002	0,0050671
prime	-0,113	0,076006	-1,49	0,145	-0,2666168	0,0406102
female#prime	0,078902	0,115464	0,68	0,498	-0,1544582	0,3122625
educ	-0,01471	0,023184	-0,63	0,529	-0,0615623	0,0321493
overconf	-0,00385	0,007225	-0,53	0,597	-0,018448	0,0107545
finlit	0,03183	0,022403	1,42	0,163	-0,0134475	0,0771076
_cons	0,933912	0,39461	2,37	0,023	0,1363761	1,731448
Statistic	Value					
Dependent var	belief_index_core					
Number of obs	47					
F-statistic	1,76					
Prob > F	0,1244					
R-squared	0,2397					
Adj R-squared	0,1032					

Variable	Coef	StdErr	t	P> t	[95%_low]	[95%_high]
female	-0,3492254	0,113774	-3,07	0,004	-0,57936	-0,1190949
prime	0,0072775	0,09376	0,08	0,939	-0,18237	0,1969251
female#prime	0,2425806	0,14079	1,72	0,093	-0,04219	0,5273557
educ	0,0244054	0,026316	0,93	0,359	-0,02882	0,0776343
overconf	0,0018891	0,008275	0,23	0,821	-0,01485	0,0186274
exper	0,0259914	0,013465	1,93	0,061	-0,00124	0,0532271
finlit	0,0270277	0,028401	0,95	0,347	-0,03042	0,0844735
_cons	0,0323296	0,453862	0,07	0,944	-0,88569	0,9503531
Statistic	Value					
Dependent var	risky_share_hyp					
Number of obs	47					
F-statistic	5,59					
Prob > F	0,0002					
R-squared	0,5009					
Adj R-squared	0,4113					

Variable	Coef	StdErr	t	P> t	[95%_low]	[95%_high]
female	-0,1824233	0,073156	-2,49	0,017	-0,33028	-0,0345697
prime	-0,0977641	0,062681	-1,56	0,127	-0,22445	0,0289196
female#prime	0,0798665	0,095222	0,84	0,407	-0,11258	0,2723167
educ	0,0101127	0,019119	0,53	0,6	-0,02853	0,0487543
overconf	-0,0055756	0,005958	-0,94	0,355	-0,01762	0,006466
finlit	0,0165903	0,018475	0,9	0,375	-0,02075	0,0539303
_cons	0,5101319	0,325431	1,57	0,125	-0,14759	1,167852
Statistic	Value					
Dependent var	belief_index_ext					
Number of obs	47					
F-statistic	2,34					
Prob > F	0,0428					
R-squared	0,2958					
Adj R-squared	0,1694					

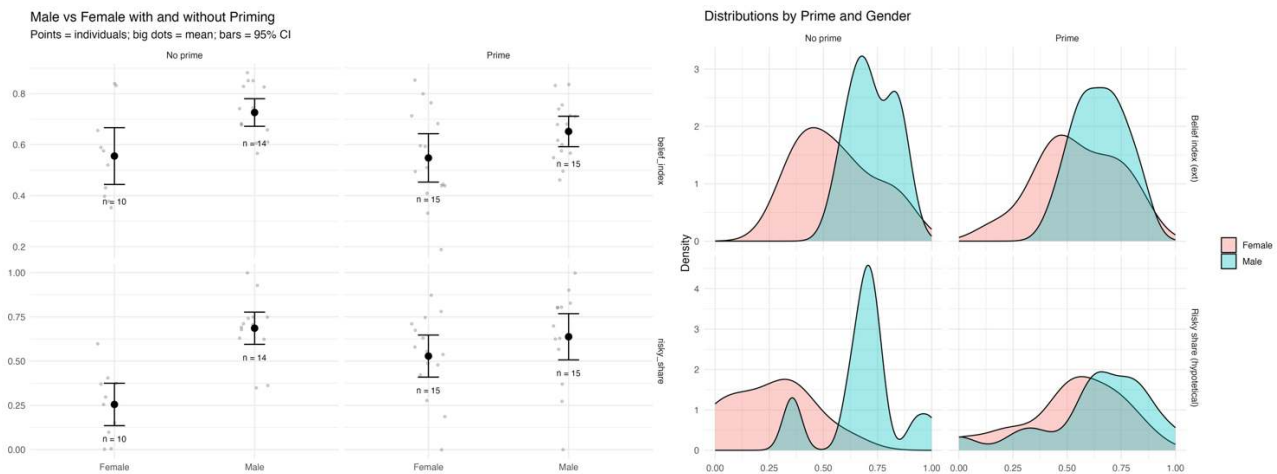
Statistically significant coefficients are highlighted in yellow, control variables in orange, and the dependent variable in green.

The regression on the actual risky share shows that none of the control variables are statistically significant (the only statistically significant variable is the constant, significant at the 10% level). This is probably due to the reduced sample size, as this index uses only 36 observations (11 fewer observations compared to the other regressions). The dataset is smaller because not all the individuals who participated in the survey actually own financial assets. The explanatory power is very low (14%).

The regression on the hypothetical risky share reveals several interesting findings. The coefficient on female is large, negative, and strongly significant, confirming that female respondents allocate a significantly smaller share of their hypothetical portfolio to risky assets compared to males. Experience is also statistically significant at the 10% level (close to 5%) and is positively associated to the risky share (individuals who are more experienced tend to have a higher risky share); The interaction term between gender and the prime is marginally significant (at the 10% level) and suggests that the treatment effect for women is less negative than the effect for men; however, the combined effect of the prime on women remains small and statistically weak. The coefficients for prime and overconfidence are not statistically significant. Education is not statistically significant as well, however its non significance is more reasonable since most students who took the survey are in between the end of the bachelor program and the end of the master program. The R squared is high (50%), which suggests that this regression has high explanatory power. These results are consistent with what we expected.

Belief_index_core and belief_index_ext show very similar results. The only statistically significant (and relevant) variable that we can see is in Female; the coefficient is negative and is statistically significant almost at the 5% level in the first index and almost at the 1% level in the second index, which is consistent with the well documented finding that women tend to exhibit lower risk tolerance than men.

Gender differences in beliefs and risk taking:



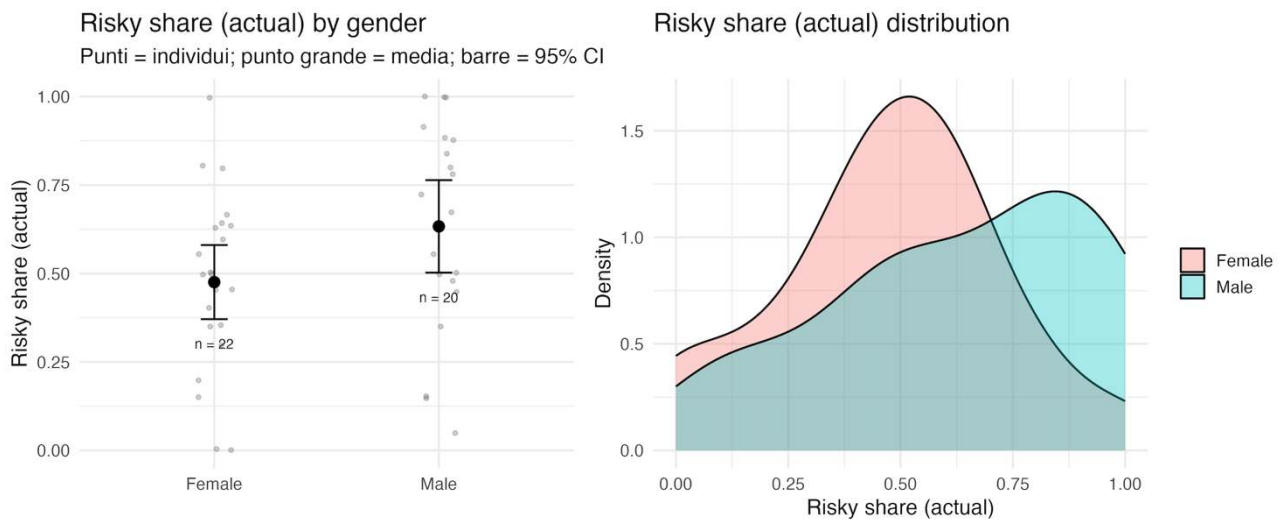
Across both visualizations a clear pattern is shown: male respondents consistently report higher belief scores compared to female respondents, with narrower confidence intervals. The density plots further show that male respondents are shifted to the right relative to female participants and are more concentrated, indicating stronger and homogeneous agreement in beliefs about gender differences. This means that male respondents perceive a stronger gender difference in risk taking compared to female participants. The prime seems to shift down belief index of male respondents by a small (non significant) amount, while it doesn't show relevant changes for female participants.

The pattern for the risky share index is similar and is also dominated by gender. In both treatment and control groups, the density plot shows that men choose substantially riskier hypothetical portfolios than

women, with male responses concentrated around higher risky share values. We can see that the prime increases the risky share of female participants (they tend to invest in more risky assets when they see the prime statement); however, due to the large confidence intervals, this change is not statistically significant.

Taken together, the two figures show strong and consistent gender differences in both beliefs and risk taking across all conditions. Women consistently display lower belief scores and lower risky share levels than men. The priming intervention, however, does not meaningfully alter these patterns.

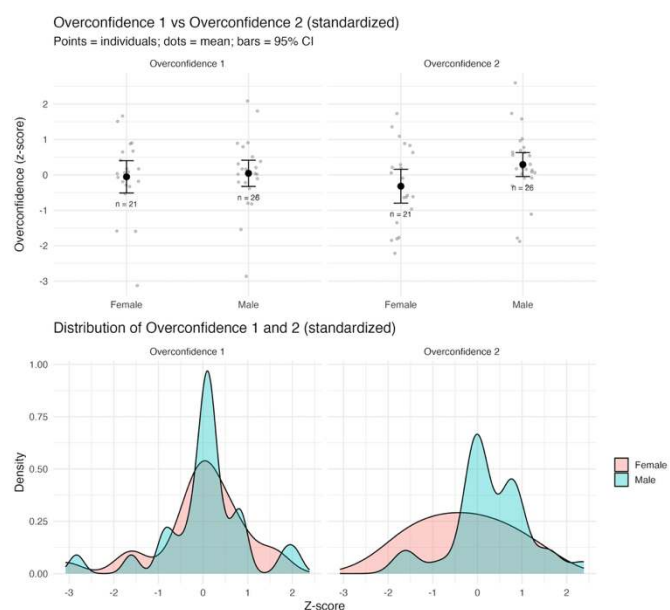
Risky share in portfolio composition:



One could argue that these results are not robust because the hypothetical task involves allocating “imaginary” savings into financial assets. The figure shows a pattern that closely mirrors the one observed in the hypothetical risky share allocation: male participants consistently display higher risky share levels than female participants. The distributions confirm that men hold portfolios with a larger share of risky assets, while women tend to hold safer portfolios. This suggests that the gender gap observed in the hypothetical scenario is also present in real financial behaviour.

Overconfidence indexes:

As previously described, we constructed two measures of overconfidence. The plot and the distribution on the left show the standardized overconfidence level based on the financial literacy questions, while the graph and the distribution on the right show the standardized overconfidence level based on the expected annual return. On average male and female are not overconfident and can predict pretty well their correct answers (although male participants display a few extreme outliers in both directions). However, when asked whether they can outperform the market, male participants are more likely to be overconfident.



Overall, these results suggest that overconfidence manifests differently depending on how we are measuring it: respondents are generally well calibrated about their own knowledge, but men display greater overconfidence when forming expectations about future market performance.

Conclusion and reflection:

Our study reveals patterns that closely mirror the existing literature. There is a significant difference both in the perceived difference between genders and in the actual risk taking in investment decisions. It's interesting to notice that these differences persist even among students; pursuing an economic/finance degree does not eliminate the documented gender disparities in financial decision making.

The prime statement doesn't have any significant effect on the treatment group that behaves very similarly to the control group. The stability of these patterns suggests that these perceptions are deeply rooted and that preferences are not easily altered through brief informational cues.

While working on the project we thought, of some ways to, increase the robustness of our results:

- Obtaining more observations by asking more people to take part of the survey
- Incentivise the participation using prizes (someone, randomly chosen, can win money)
- Overconfidence measure: in this project we used two proxies to measure overconfidence. Having two different overconfidence indexes was useful for the goal we had in mind, however creating a unique index by combining both could have strengthened results
- Factor analysis: this technique could be applied both to derive a unified overconfidence index and to validate and refine the weights used to construct the risky share indexes (similarly what Lusardi, van Rooij and Alessie did in "Financial Literacy and Stock Market Participation")
- Bootstrapping: while it does not increase the number of observations, bootstrapping could improve the reliability of estimates, especially with a relatively small sample, by providing robust confidence intervals through repeated resampling.
- Measuring ability, numeracy sophistication; several studies show that individuals with stronger mathematical skills are more likely to own stocks. Similarly, it is important to account for social networks and peer effects, which can also shape investment behaviour.

Exam questions:

Having a mismatch between the unit of randomization and the unit of analysis is something that the researchers should take into account. The main problem is that the number of observations becomes artificially higher, since there are several students for each class. Moreover, students are not independent units of observation since they share many characteristics such as professors or physical environment. The main issue lies not with the coefficient estimation, but with the standard errors. Standard errors are computed using the following formula:

$$SE_{\bar{x}} = \frac{s}{\sqrt{n}} \quad \begin{array}{l} s = \text{Standard deviation} \\ n = \text{Sample size} \end{array}$$

Increasing the sample size causes S.E. to be significantly smaller. This can alter the significance of the coefficients in our regression. Consequently, we might incorrectly conclude that the treatment has a

statistically significant effect simply because the independence of observations was violated. To solve this issue we could use aggregate data (sum of test scores) or cluster robust standard errors. Moreover, there is another problem in scaling up to nationwide program with this setting: self selection bias; since the schools are free to decide whether to participate in the experiment or not, we will end up with a convenience sample rather than a representative one. When scaling up, effects may shrink because schools are heterogeneous. This problem could have been solved ex-ante by selecting schools in a way that ensures representativeness of the population distribution.

Our goal is to determine what drives the demand for these super short loans. In particular we would like to understand whether the preference for these types of loans is determined by standard preferences or behavioural failures. We could request administrative data to track customers over time (look at hard information from company and background data below); after collecting this data, a good approach could be constructing a survey that could allow us able to identify these preferences and identify users. We would like to be able to see whether these individuals are financially constrained or not, obtain data about their demographics, understanding the reason why they want the loan (income shocks/necessities) and what drives their decision. After collecting all this data we could use a similar approach to the paper of A. & T. to see whether informed individuals would still take the loan or not. An approach to do that would be implementing a survey with the following structure:

Part I: Background Data

(The following variables are collected prior to the experiment)

1. Demographic Data

- Gender, Age, Education Level, Employment Status.

2. Financial State

- Monthly Income, Existing Debts, Credit Score, Past use of payday loans.

3. Financial Knowledge

- Understanding of Interest Rates, Inflation and other key economic concepts.

4. Time Preferences

- Discounting check (Money today vs. Money in future).

Part II: Information Treatment

Participants are randomly assigned to read BOX A or BOX B.

BOX A: CONTROL GROUP (Standard Info)

These loans are designed to provide immediate cash for short-term needs.

You will receive the principal amount today and must repay the principal plus the fee by the deadline.

Please review the amount, fee, and deadline for each offer below before accepting.

BOX B: TREATMENT GROUP (Risk Info)

These loans are designed to provide immediate cash for short-term needs.

You will receive the principal amount today and must repay the principal plus the fee by the deadline.

Please review the amount, fee, and deadline for each offer below before accepting.

ADDITIONAL INFORMATION:

Research shows that using high-interest loans significantly reduces your wealth because the interest costs eat into your income.

Failure to repay these loans on time can severely damage your credit score, reducing your probability of obtaining a mortgage, car loan, or credit card in the future.

Part III: Loan Decisions

Based on the information provided, would you Accept or Reject the following offers?

SET 1: Varying Interest Rate (Constant Duration: 2 Weeks)

Loan	Fee	Implied Rate	Decision	
\$500	\$25	5%	☐ Yes	☐ No
\$500	\$50	10%	☐ Yes	☐ No
\$500	\$75	15%	☐ Yes	☐ No
\$500	\$100	20%	☐ Yes	☐ No
\$500	\$150	30%	☐ Yes	☐ No

SET 2: Varying Duration (Constant Fee Rate: 20% of Principal)

Loan	Total Fee	Due In	Decision	
\$500	\$100	1 Week	☐ Yes	☐ No
\$500	\$100	2 Weeks	☐ Yes	☐ No
\$500	\$100	1 Month	☐ Yes	☐ No
\$500	\$100	3 Months	☐ Yes	☐ No

Hard information from company

- Loan history: amounts, dates, duration, rollovers, late payments, defaults.
- Income and pay schedule: wage level, pay frequency, paycheck dates.
- Other credit: credit cards, overdrafts, other loans and credit limits.
- Future outcomes: repeat borrowing, subsequent defaults, early repayment.

Extensions of the survey could address:

- probability of rolling over;
- over-optimism about future self-control;
- role of education, wealth and other key factors

This is an example of a survey structure that could be used to distinguish preferences and test the knowledge of the participants. Combining the firm's administrative data with this survey and the information treatment lets us separate preferences from mistakes. If, once fully informed about true costs and risks, customers' borrowing choices remain similar, this suggests rational demand under tight liquidity constraints, otherwise misperceptions and behavioural biases as the main drivers of borrowing.

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